3D Brain Tumor Segmentation

Pretrained UNet3D and UNETR on real BraTS-Africa MRI

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Public project: https://github.com/ZekRock18/brain-tumor-segmentation

What this project demonstrates

Two released pretrained neural networks segment tumor regions in real three-dimensional brain MRI. The completed Kaggle run processes four patients with both models, compares their predictions with supplied segmentation labels, and exports numerical results, comparison figures and animated slice sequences. No training or fine-tuning was performed.

Execution	Verified outcome
Platform	Kaggle version 6; Tesla T4 GPU; CUDA inference
Coverage	4 patients × 2 models = 8 model-case evaluations
Evidence	15 executed code cells; no notebook execution errors; completion marker and 19 expected report artifacts
Checkpoint integrity	Both models loaded strictly; checkpoint SHA-256 hashes recorded

Measured results from this run

Dice measures region overlap (higher is better). HD95 measures boundary distance in millimetres (lower is better). Values below are means across the four patients and, for the mean columns, the three tumor regions.

Model	ET Dice	TC Dice	WT Dice	Mean Dice	HD95 mm
UNet3D	0.8611	0.7868	0.9158	0.8546	6.3594
UNETR	0.8618	0.8281	0.8788	0.8562	5.6408

These are our saved inference results, not the model author's validation scores. The difference in mean Dice is small; four cases cannot establish a general model ranking.

Method and scope

1. Real MRI data

The attached BraTS-Africa dataset yielded 146 complete cases during discovery. A seeded sample (seed 42) selected four cases, comprising two in the Glioma cohort and two in Other Neoplasms. This composition arose from sampling; it was not a stratified study. Each case contains T1, contrast-enhanced T1, T2, FLAIR and a reference segmentation. The sampled image tensors are 4 × 240 × 240 × 155.

2. Preprocessing and label conversion

Each modality is normalized using the repository's positive-intensity brain mask, 0.5th/99.5th percentile clipping and brain-only z-score normalization. Background voxels are set to zero. The implementation remaps label 3 to 4, then creates overlapping masks: enhancing tumor (ET) = 4; tumor core (TC) = 1 or 4; whole tumor (WT) = 1, 2 or 4.

The mapping follows the repository convention. Independent confirmation of annotation semantics against dataset documentation remains a limitation; the scores should be read with that qualification.

3. Released pretrained models

UNet3D uses a convolutional encoder-decoder with skip connections; UNETR combines a transformer encoder with a convolutional decoder. The architectures and checkpoints come from the credited original project. Strict state-dictionary loading passed for 19,992,603 UNet3D parameters and 19,327,907 UNETR parameters.

4. Full-volume inference

Each model predicts on overlapping 128 × 128 × 128 patches, with 25% overlap and Gaussian blending. Sigmoid outputs are thresholded at 0.5. Evaluation mode and disabled gradients prevent learning. CUDA mixed precision is used on the GPU. Metrics are computed on complete three-dimensional volumes, not only the displayed slices.

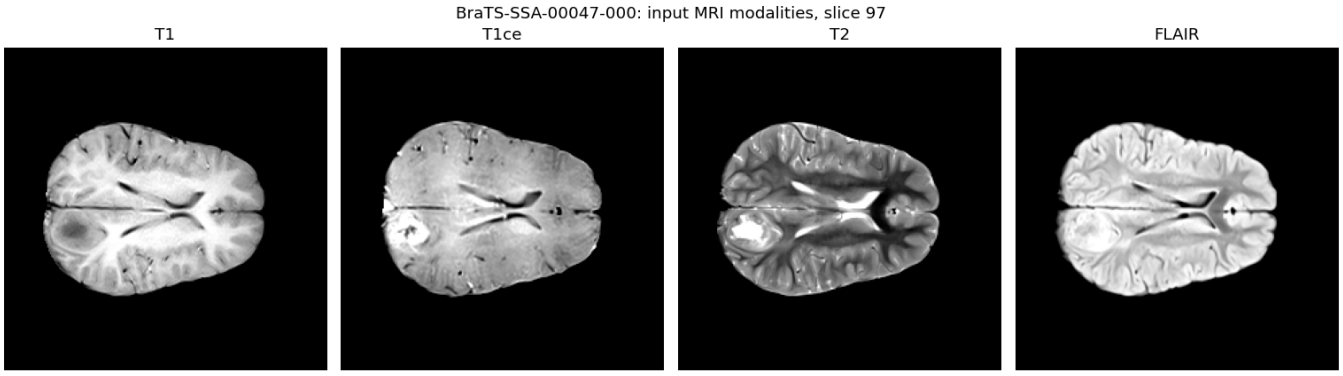
5. Evidence and deliverables

The saved executed notebook, per-region CSV, per-cohort CSV, checkpoint hashes and execution log support the reported run. Four comparison figures, four modality previews and four animated slice sequences accompany an offline HTML report. The animations traverse axial slices through each volume; they are not 3D surface renderings.

From MRI to segmentation



Workflow diagram: the same normalized input is evaluated by both pretrained models. The reference label is used for scoring and visualization, not to update model weights.



Reading the four input images

These are the actual modality previews for BraTS-SSA-00047-000. They show different contrasts of the same patient anatomy. The input channels are combined by the model; the tumor prediction is not obtained by simply coloring bright pixels. The supplied reference segmentation is a separate file used for comparison.

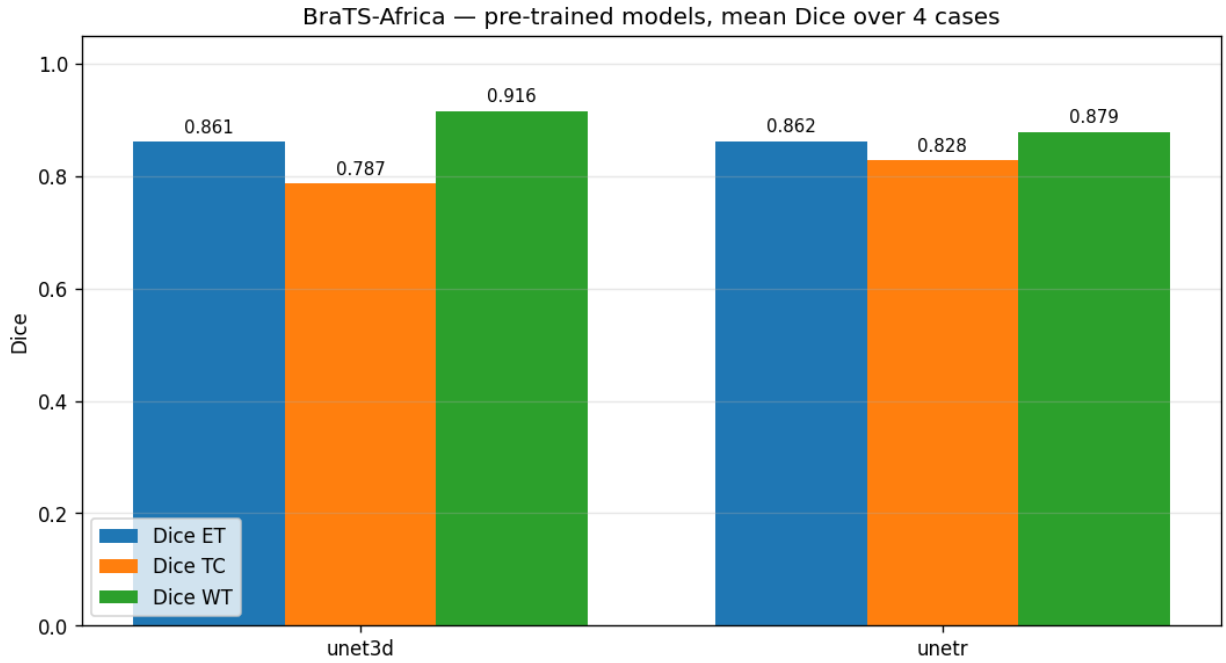
Why the output is useful to inspect

A label map assigns a predicted region membership to each voxel. The overlays make the spatial agreement and missed or excess regions visible. The quantitative evaluation summarizes full-volume agreement, while the images help explain individual cases. Neither replaces a clinical assessment.

Results and interpretation

Case suffix	Cohort	UNet3D Dice	UNETR Dice	UNet3D HD95	UNETR HD95
00047-000	95_Glioma	0.9105	0.8606	1.93	4.79
00068-000	95_Glioma	0.7635	0.7797	6.20	7.38
00111-000	51_OtherNeoplasms	0.8831	0.8465	3.13	3.63
00170-000	51_OtherNeoplasms	0.8611	0.9381	14.18	6.75

Every entry is a mean over ET, TC and WT for that patient. HD95 is in millimetres. Full case IDs begin with BraTS-SSA-.



What we can conclude

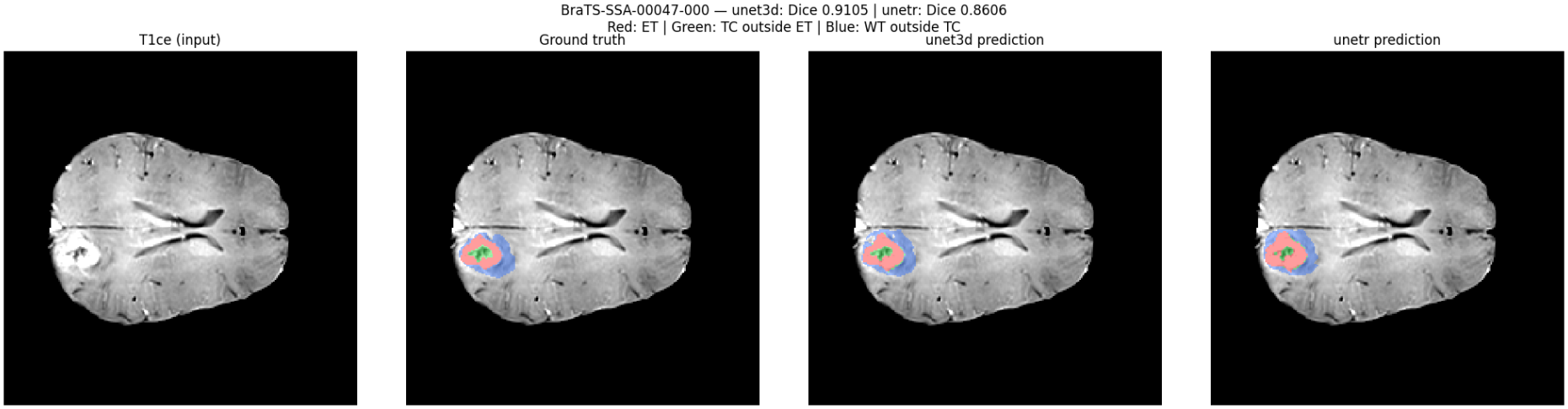
Both pretrained models produced full-volume segmentations on all four real patients. UNETR has slightly higher mean Dice in this sample and lower overall mean HD95; each model has higher mean Dice on two patients. UNet3D has higher average WT Dice. Region-specific and patient-specific differences matter more than declaring an overall winner from this small sample.

Limits and attribution

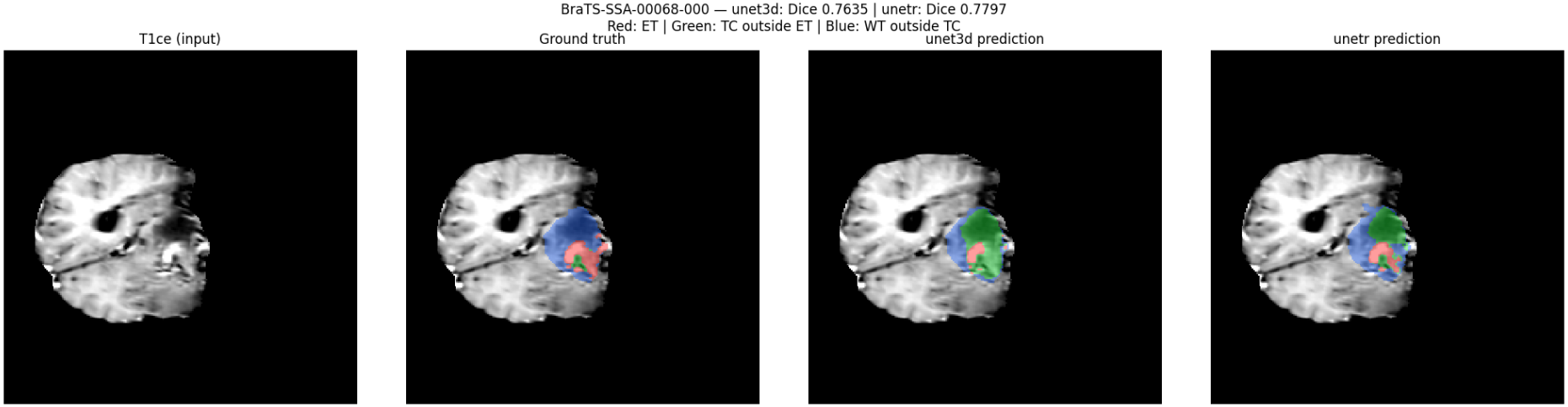
This is a small illustrative evaluation, not a clinical validation or reproduction of the original training benchmark. Training-set overlap has not been independently excluded, so the patients are not claimed to be guaranteed unseen. No training curves, fine-tuning gains or peak-memory results are claimed. Our contribution is the Kaggle integration, data adaptation, evaluation, visual reporting and execution verification; model design and pretrained training belong to the original authors.

Patient comparisons 1–2

Left to right: T1ce MRI, reference segmentation, UNet3D prediction, UNETR prediction. Red: ET; green: core outside ET; blue: whole tumor outside core. Displayed slice has maximal reference WT area. Metrics describe the full volume.



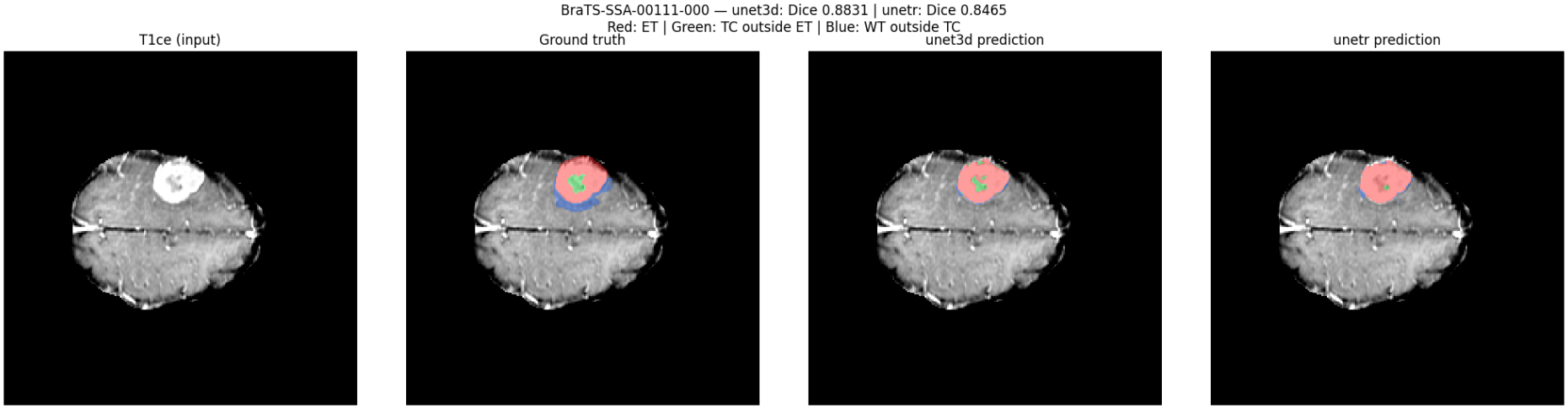
BraTS-SSA-00047-000 | See results/figures/BraTS-SSA-00047-000.gif for the axial slice animation.



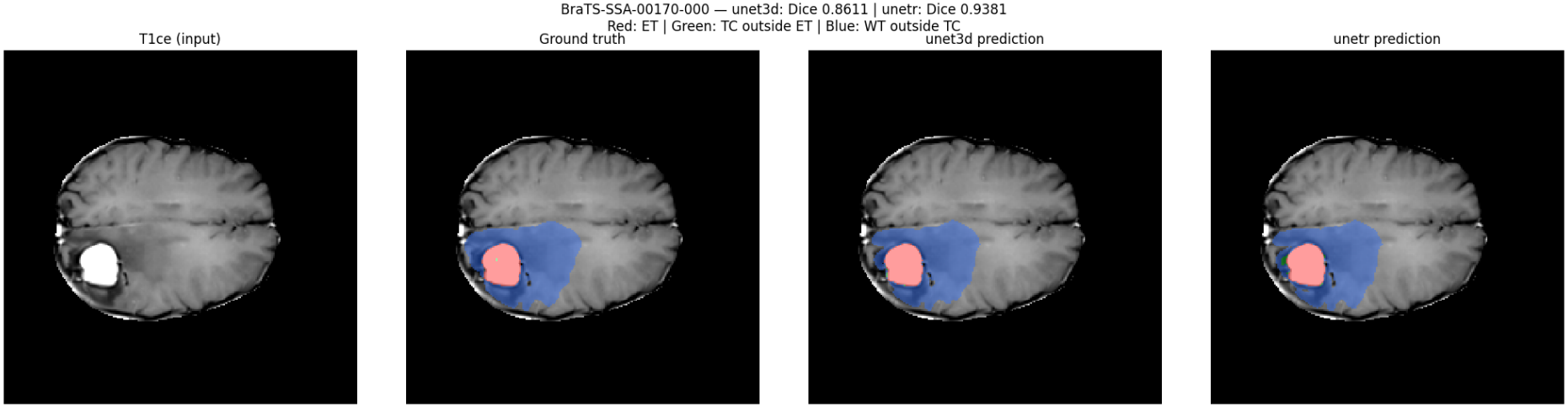
BraTS-SSA-00068-000 | See results/figures/BraTS-SSA-00068-000.gif for the axial slice animation.

Patient comparisons 3–4

Left to right: T1ce MRI, reference segmentation, UNet3D prediction, UNETR prediction. Red: ET; green: core outside ET; blue: whole tumor outside core. Displayed slice has maximal reference WT area. Metrics describe the full volume.



BraTS-SSA-00111-000 | See results/figures/BraTS-SSA-00111-000.gif for the axial slice animation.



BraTS-SSA-00170-000 | See results/figures/BraTS-SSA-00170-000.gif for the axial slice animation.

Full regional results

Case suffix	Model	ET Dice	TC Dice	WT Dice
00047-000	UNet3D	0.8971	0.9381	0.8964
00047-000	UNETR	0.8561	0.9082	0.8174
00068-000	UNet3D	0.7023	0.6352	0.9529
00068-000	UNETR	0.7722	0.6393	0.9276
00111-000	UNet3D	0.8782	0.9120	0.8592
00111-000	UNETR	0.8383	0.8693	0.8319
00170-000	UNet3D	0.9668	0.6618	0.9548
00170-000	UNETR	0.9807	0.8953	0.9384

Case suffix	Model	ET HD95	TC HD95	WT HD95
00047-000	UNet3D	1.56	1.41	2.83
00047-000	UNETR	2.45	2.24	9.70
00068-000	UNet3D	4.69	12.17	1.73
00068-000	UNETR	3.00	12.45	6.70
00111-000	UNet3D	2.83	2.83	3.74
00111-000	UNETR	3.32	3.46	4.12
00170-000	UNet3D	13.63	26.66	2.24
00170-000	UNETR	1.00	16.25	3.00

HD95 distances are in millimetres. Values are rounded for presentation; results/brats_africa_results.csv contains full precision. Every case and both models are included, not only the strongest examples.

Sources and evidence

Where to inspect the completed run

Open results/report.html for the offline visual report. Open results/executed.ipynb to inspect the actual outputs of all executed cells. Use brats_africa_results.csv for full-precision case-by-model metrics and run_metadata.json for settings, hardware and checkpoint identities. The clean notebook in notebooks/ is for rerunning, not evidence of a new execution.

Project owner: Prakhar Gupta (ZekRock18). Project team: Prakhar Gupta and Aditya Dubey. Our work covers the Kaggle integration, data adaptation, pretrained evaluation, visual reporting and verified delivery. Original model architectures and pretrained checkpoints are credited separately.

Our project on GitHub — Prakhar Gupta / ZekRock18
https://github.com/ZekRock18/brain-tumor-segmentation

Public Kaggle notebook (packaged results: version 6)
https://www.kaggle.com/code/zekrock/brain-tumor-seg-brats-africa

Original model repository
https://github.com/MohanadMahran/Brain_Tumor_Benchmark

UNet3D checkpoint
https://huggingface.co/MohanadKombar/brain-tumor-unet3d-brats

UNETR checkpoint
https://huggingface.co/MohanadKombar/brain-tumor-unetr-brats

BraTS-Africa input dataset
https://www.kaggle.com/datasets/yakunokamizen/brats-africa

Attached checkpoint dataset
https://www.kaggle.com/datasets/zekrock/brats-weights

Ownership and data handling

Original model code and checkpoint authors are credited through the links above. This delivery does not assign a new license to their artifacts. Retain upstream attribution and follow the respective source and dataset terms. Raw MRI inputs are not redistributed in this folder; they remain attached on Kaggle. Figures are generated outputs from the selected cases.